

Ethan Tran

ethantran@quwin.dev | github.com/quwin | linkedin.com/in/kien-ethan-tran | www.quwin.dev

SUMMARY

Applied AI software engineer building and evaluating model-backed systems with Python, PyTorch, C++, and CUDA. Experienced turning ambiguous requirements into reliable pipelines, APIs, and performance-oriented software.

EDUCATION

California State University, Sacramento

Bachelor of Science, Computer Science – GPA: 3.47

Expected December 2027

EXPERIENCE

Software Engineer Intern

July 2026

Synk

- Designed pipeline project for generating low-poly avatar concepts for Synk's browser-based 3D career fairs and networking environments, focusing on the key requirements of professional presentation and distinctive visual identity.
- Evaluated multiple image-generation frameworks for output consistency, visual clarity, prompt adherence, and suitability in production across multiple design targets.
- Scoped designs for feasible avatar Three.js implementations using simple geometric components and emissive displays.

Software Engineer Extern — Pfizer-hosted Externship

June 2026

Extern

- Built an OCR-backed document pipeline with Python, PyTorch, and Tesseract to extract, classify, and route content from 1,000+ page pharmaceutical files.
- Partnered with Pfizer stakeholders to translate document-routing pain points into pipeline requirements and evaluation criteria.
- Improved document retrieval accuracy by 36% through systematic tuning of segmentation, indexing configuration, and model selection.
- Delivered a comprehensive technical report that outlined model trade-offs, optimization strategies, and deployment recommendations, enabling stakeholders to make informed decisions on the system rollout.

PROJECTS

GitHub Documentation RAG Service | *Python, FastAPI, Qdrant, Docker, Google Cloud*

- Built a modular RAG service spanning GitHub ingestion, metadata-aware chunking, hybrid retrieval, cross-encoder reranking, grounded generation, and answer evaluation.
- Implemented configurable hybrid search combining dense embeddings, sparse retrieval, and cross-encoder reranking for complex documentation queries.
- Deployed typed API endpoints returning source-backed answers, validity, confidence, and completeness signals.
- Evaluated multiple indexing and ranking strategies, improving answer grounding to 92% faithfulness and 99% citation accuracy on a 50-question evaluation suite.

GPU-Accelerated Postflop Poker Solver | *C++, CUDA, PyTorch, FLTK*

- Developed and deployed a custom No-Limit Hold'em postflop solver using Counterfactual Regret Minimization (CFR) across configurable game trees.
- Implemented CFR+ and exploitability evaluation to iteratively compute and assess equilibrium poker strategies.
- Parallelized solver computation with CUDA and memory-aware batching to support large workloads on consumer GPUs.

Distributed API Gateway + Rate Limiter | *Go, Lua, Nginx, Redis, Prometheus, Docker, Google Cloud*

- Engineered a Go gateway with API-key authentication, per-principal throttling, reverse-proxy routing, quota headers, and explicit failure handling.
- Implemented fixed-window, token-bucket, and sliding-window limiters with in-memory and Redis backends, using atomic Lua scripts for shared replica state.
- Authored deterministic tests for timing boundaries, refill and expiry behavior, principal isolation, and state transitions.

SKILLS

Languages: Python, C++, Go, TypeScript, JavaScript, SQL

AI / ML: PyTorch, Tesseract, LangChain, Hugging Face, TensorFlow, Scikit-learn, Claude Dev.

Backend / Systems: Google Cloud, AWS, Docker, FastAPI, Qdrant, Redis, PostgreSQL, Nginx, Git

Frontend / Frameworks: React, Next.js, Node.js, Tailwind CSS, Flask, CUDA